221-2327 M. Rohaan A# 02:-

Q1:- Sol.

a. Casual vs Non-Casual System

Casual:- Output at time t depends only on present and past inputs (not future). Formally, $y(t_0)$ depends only on $x(\tau)$ for $\tau \leq t_0$.

Non-Casual:- Output depends on future values of the input (i.e needs $x(t+\Delta)$ for $\Delta > 0$).

Example

Casual :- $y(t) = x(t) + 2x(t-1)$ — depends on present and past only.

Non-Casual :- $y(t) = x(t+1)$ — depends on future input, non-casual

Why it matters

* Physical systems are casual. Non-casual systems appear in offline signal processing

b. Time-variant vs Time-invariant System

Time-Invariant:- if input $x(t) \rightarrow y(t)$, then a time shifted input $x(t-t_0)$ produces a time shifted output $y(t-t_0)$. System properties don't change with time.

Time-Variant:- System response changes over time. A time shift of input doesn't produce the same shift in output.

Example

Time-invariant:- $y(t) = 3x(t) + x(t-2)$ — Shift input by t_0 leads to the same shift in output.

Time-variant:- $t \cdot x(t)$ — if you shift the input, the multiplicative factor t changes and the relation is not simply shifted.

*Why it matters

LTI systems have powerful tools. Time invariant ~~method~~ system require other methods.

C. Stable vs Unstable system

Stable: For every bounded input $|x(t)| < M_x$, the output is bounded $|y(t)| < M_y$.

Unstable: A bounded input can produce an unbounded output.

Example

Stable $H(s) = \frac{1}{s+2}$ (pole at $-2 \rightarrow$ stable)

Unstable $H(s) = \frac{1}{s-1}$ (pole at $+1 \rightarrow$ output grows exponentially for step input.)

*How to check for LTI systems

Continuous-Time: - all poles must have negative real parts for stability.

Discrete-Time: - all poles must lie inside the unit circle.

CamScanner

Q2:- Sol.

a. V_{in} → series R → node V_{out} → capacitor C

$$Z_r = R$$

$$Z_c = \frac{1}{sC}$$

$$V_{out} = V_{in} \frac{Z_c}{Z_R + Z_c}$$

$$H_a(s) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{1 + sRC}$$

first order low-pass with DC gain 1

$$\omega_c = \frac{1}{RC}$$

b. $V_{in} \rightarrow$ across $C \rightarrow$ node $V_{out} \rightarrow$ resistor R

$$Z_c = \frac{1}{sC}$$

$$Z_r = R$$

$$H_b(s) = \frac{V_{out}}{V_{in}} = \frac{Z_r}{Z_r + Z_c} = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{1+sRC}$$

$\omega_c = \frac{1}{RC}$ ∴ first order high-pass with DC gain 0, high-frequency gain 1.

Q3:- Sol.

$$H(s) = \frac{0.2s^2}{s^3 + 2s - 0.9}$$

Zeros :- two zeros at $s=0$

Poles :- the denominator roots

$s \approx -0.2072719 \pm j1.4594231$ (Complex)

$s \approx +0.41441439$ (Real)

Pole with positive part means system is unstable.

